Practice Quiz: Communication (Mathematical Frameworks)

Name: __ ***Date:*** __

Part A: Multiple Choice

1. Communication in Active Inference is formalized as: A) One agent sending information to a passive receiver B) Coupled inference where each agent's actions become the other's observations C) Random information exchange D) A single agent talking to itself

2. Generalized synchrony between agents is measured by: A) The speed of data transfer B) Mutual information $I(s_A; s_B)$ between internal states — positive means aligned C) The number of words exchanged D) Physical proximity

3. In the coupled free energy $F_{total} = F_A + F_B + F_{coupling}$: A) $F_{coupling}$ is always zero B) $F_{coupling}$ ensures both agents' models converge toward shared beliefs C) Only F_A matters D) F_{total} is minimized by not communicating

4. Words in language compress: A) Physical objects B) Hidden state representations from the generative model C) Neural activity directly D) Nothing meaningful

5. Cultural norms are formalized as: A) Laws enforced by police B) Shared prior distributions (D vectors) that reduce collective prediction error C) Individual preferences D) Arbitrary rules

6. Miscommunication occurs when: A) Agents speak too loudly B) Agents have significantly different generative models, causing large prediction errors C) Agents are physically far apart D) Agents speak the same language

7. Shannon's information theory treats communication as signal transmission. Active Inference treats it as: A) Exactly the same B) Coupled inference — agents actively model each other's beliefs and update jointly C) Irrelevant to humans D) Pure noise

Part B: Short Answer

1. Two people discuss whether to go to the beach. Person A has a strong prior for "beach is fun" (P(good) = 0.9). Person B has a strong prior for "beach is not fun" (P(good) = 0.2). After conversation, they reach P(good) = 0.6 and P(good) = 0.5. Did generalized synchrony increase? Show reasoning.

2. Explain how the sentence "Don't forget to pick up milk on your way home" compresses a generative model. What hidden states, observations, transitions, and preferences does it encode?

3. Why do shared cultural norms (like "drive on the right") reduce collective prediction error for all agents in a society? What happens at the collective free energy level when someone violates the norm?